

Probability Practice

September 2026

I have a small satchel that contains dice; the first set I ever bought myself and a heavy, metal set. The contain has a total of 14 dice in it. Some dice are plastic and some dice are metal. Some dice are orange and some dice are not. The below table gives the counts. Please give both the notation (e.g. $P(\text{Orange}) = .2$) and the numeric answer.

	Metal	Plastic
Orange	1	7
Blue	2	0
Other	4	0

1. If we randomly draw a dice, what is the probability it is not orange?

Probability of not orange = $P(\text{Orange}^c) =$

$$\frac{\text{Blue or Other}}{\text{Total Dice}} = \frac{6}{14} = \frac{3}{7}$$

2. If we randomly draw a dice, what is the probability it will be metal?

Probability of metal = $P(\text{metal})$

$$\frac{7}{14} = \frac{1}{2}$$

3. If we randomly draw a dice, what is the probability we draw a dice that is orange OR is plastic?

So everything that is orange and everything that is plastic = $P(\text{orange} \cup \text{plastic}) =$

$$\frac{1+7}{14} = \frac{4}{7}$$

Alternatively, we can use the additive rule...

$$P(\text{orange} \cup \text{plastic}) = P(\text{Orange}) + P(\text{Plastic}) - P(\text{Orange} \cap \text{Plastic})$$

$$\begin{aligned} P(\text{orange} \cup \text{plastic}) &= \frac{8}{14} + \frac{7}{14} - \frac{7}{14} \\ &= \frac{8}{14} \\ &= \frac{4}{7} \end{aligned}$$

4. If we randomly draw a dice, what is the probability it is both not blue AND not metal?

$$P(\text{Not Blue and Not Metal}) = P(\text{Blue}^C \cap \text{Metal}^C) =$$

$$\frac{7}{14} = \frac{1}{2}$$

This question is basically asking what is the probability we draw an (orange or other colored) dice that is (plastic)....that is things that are both not blue and not metal. Interestingly, there are no blue plastic dice so here the question actually reduces to the probability you draw a plastic dice.

5. If we randomly draw a dice and that dice is a metal, what is the probability that dice is blue?

Given we are only looking at the metal subpopulation of dice, what is the probability we draw a blue =

$$P(\text{Blue} \mid \text{Metal}) =$$

$$\frac{2}{7}$$

6. Difficult: What is the highest probability that drawing a blue or green dice can have? That is, what is the upper limit for the probability of drawing a blue or green dice.

So the trick to this question is realizing how many dice *could* be green.....The highest possible number would be 4 where all four dice colored as “Other” are actually green. So for the problem we do assume it’s 4. This creates an *upper bound* on the probability we draw and blue or green dice and is a common strategy in statistics. Sometimes it’s hard to give the exact theoretical probability but we can give boundaries around it that are more easy to calculate.

$$P(\text{Blue} \cup \text{Green}) \leq P(\text{Blue}) + P(\text{Green})$$

$$\frac{2}{14} + \frac{4}{14} = \frac{6}{14}$$